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Inventor(s)	Ballantine; Arne et al.

Power source communication with downstream device by changing output voltage on a power bus

Abstract

A power generation system includes a power source that is configured to communicate with at least one of a downstream load or a downstream device by changing a voltage on a power bus between the power source and the at least one of the downstream load or the downstream device, while power source provides power on the power bus to the at least one of the downstream load or the downstream device.

Inventors: Ballantine; Arne (Palo Alto, CA), Pmsvvsv; Prasad (Sunnyvale, CA), Gopinath; Rajesh (Sunnyvale, CA), Gurunathan; Ranganathan (Bangalore, IN), Harris; Robert (Mebane, NC), Worth; Marshall (Raleigh, NC), Mandela; Chaitanya (Bangalore, IN), Gopalakrishnan; Vishal Anand (Bangalore, IN)

Applicant: BLOOM ENERGY CORPORATION (San Jose, CA); POWERSECURE, INC. (Wake Forest, NC)

Family ID: 69640261

Assignee: BLOOM ENERGY CORPORATION (San Jose, CA); POWERSECURE INCORPORATED (Wake Forest, NC)

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Primary Examiner: Douyette; Kenneth J

Attorney, Agent or Firm: THE MARBURY LAW GROUP PLLC

Background/Summary**FIELD**

(1) The present invention is generally directed to power generation systems and, in particular, to a fuel cell system that communicates with downstream devices using an output voltage droop on a power bus.

BACKGROUND

(2) Electrochemical devices, such as fuel cells, can convert energy stored in fuels to electrical energy with high efficiencies. In a fuel cell system, such as a solid oxide fuel cell (SOFC) system, an oxidizing flow is passed through the cathode side of the fuel cell while a fuel inlet flow is passed through the anode side of the fuel cell. The oxidizing flow is typically air, while the fuel flow can be a hydrocarbon fuel, such as methane, natural gas, pentane, ethanol, or methanol. The fuel cell enables the transport of negatively charged oxygen ions from the cathode flow stream to the anode flow stream, where the ion combines with either free hydrogen or hydrogen in a hydrocarbon molecule to form water vapor and/or with carbon monoxide to form carbon dioxide. The excess electrons from the negatively charged ion are routed back to the cathode side of the fuel cell through an electrical circuit completed between anode and cathode, resulting in an electrical current flow through the circuit.

(3) SOFC systems may be used to power many different devices under many different arrangements. The variety of potential applications creates a need for an SOFC system and control method that can be readily adapted for use with different kinds of systems using different control hardware and software.

SUMMARY

(4) An embodiment includes a power generation system, comprising a power source that is configured to communicate with at least one of a downstream load or a downstream device by changing a voltage on a power bus between the power source and the at least one of the downstream load or the downstream device, while power source provides power on the power bus to the at least one of the downstream load or the downstream device.

(5) An embodiment includes method of operating a power source, comprising providing output power on a power bus from a power source to at least one of a load or a device, and changing an output voltage of the power source on the power bus for the power source to communicate with the at least one of the load and the device while the power source provides the output power on the power bus.

(6) Another embodiment includes a method of operating a power source, comprising providing

output power from a power source to a load to track and meet a power demand of the load, and directing excess output power from the power source to an additional load to maintain the output power of the power source above a minimum desired operating power of the power source when the power demand from the load drops to less than the minimum desired operating power of the power source.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a fuel cell system according to embodiments.
- (2) FIG. 2 is a schematic side cross-sectional view of a hot box according to embodiments.
- (3) FIG. 3A is a schematic diagram of a system according to embodiments.
- (4) FIG. 3B is a flowchart of steps in an exemplary method for altering the output power of the system shown in FIG. 3A.
- (5) FIGS. 4A-4D show temporal plots of certain aspects of the system of FIG. 3A in operation according to an embodiment.
- (6) FIG. 5 shows temporal plot of certain aspects of the system of FIG. 3A in operation according to another embodiment.
- (7) FIGS. 6A-6C show temporal plots of certain aspects of the system of FIG. 3A in operation according to an embodiment.
- (8) FIGS. 7A-7C show temporal plots of another aspect of the system of FIG. 3A in operation according to another embodiment.

DETAILED DESCRIPTION

- (9) The present inventors realized that one way to address expanding the compatibility and applicability of power sources, such as SOFC systems, is by developing control methods/systems that are compatible with different loads or devices. In particular, the present inventors realized that signaling those loads or devices regarding output power of the power source without relying on specific protocols or control hardware in the loads or devices would be advantageous.
- (10) Embodiments include a power source that communicates with the downstream load and/or device by changing voltage on power bus between the power source and the load and/or device, while power source provides power on the power bus to the load and/or device.
- (11) In one embodiment, the power source may be a DC power source, such as a fuel cell system, a photovoltaic system, a thermoelectric system, etc. The power bus may be a DC power bus. The load may be a microgrid. The device may be an inverter containing device, such as an uninterruptible power system (UPS) (also referred to as an uninterruptible power module). The communication may be carried between separate power source controller and inverter controller which do not have a separate communication link or protocol between them.
- (12) For example, in one embodiment output power is provided from the power source to the load to track and meet a power demand of the load. In response to an increased power demand from the load, the power source controller determines if the requested increase in output power of the power source would cause the output power of the power source to exceed a threshold output power. If the output power of the power source would exceed the threshold output power, then the power source controller signals to the inverter controller that the threshold output power would be exceeded by changing (e.g., drooping) an output voltage of the power source, and does not increase the output power above the threshold output power such that the output power no longer tracks the power demand of the load. If the output power of the power source would not exceed the threshold output power, then the power source controller increases the output power from the power source to meet the power demand of the load.
- (13) In another embodiment, when the load power demand drops to less than a minimum desired

power output of the power source, such as solid oxide fuel cell system, then power is directed to an additional, non-critical load, such as a resistive load bank and/or an energy storage device, such as a battery or supercapacitor. The power directed to the additional load may be equal to a difference between the minimum desired operating power and the load power demand.

(14) FIG. 1 illustrates an exemplary modular fuel cell system described in U.S. Pat. No. 8,440,362, incorporated herein by reference in their entirety. The modular system may contain modules and components described above as well as in U.S. patent application Ser. No. 11/656,006, filed on Jan. 22, 2007, and entitled “Modular Fuel Cell System” which is incorporated herein by reference in its entirety. The modular design of the fuel cell system enclosure **10** provides flexible system installation and operation.

(15) The modular fuel cell system enclosure **10** includes a plurality of power module housings **12** (containing a fuel cell power module components), one or more fuel input (i.e., fuel processing) module housings **16**, and one or more power conditioning (i.e., electrical output) module housings **18**. For example, the system enclosure may include any desired number of modules, such as 2-30 power modules, for example 6-12 power modules. FIG. 1 illustrates a system enclosure **10** containing six power modules (one row of six modules stacked side to side), one fuel processing module, and one power conditioning module, on a common base **20**. Each module may comprise its own cabinet or housing. Alternatively, the power conditioning and fuel processing modules may be combined into a single input/output module located in one cabinet or housing **14**. For brevity, each housing **12**, **14**, **16**, **18** will be referred to as “module” below.

(16) While one row of power modules **12** is shown, the system may comprise more than one row of modules **12**. For example, the system may comprise two rows of power modules stacked back to back.

(17) Each power module **12** is configured to house one or more hot boxes **13**. Each hot box contains one or more stacks or columns of fuel cells (not shown for clarity), such as one or more stacks or columns of solid oxide fuel cells having a ceramic oxide electrolyte separated by conductive interconnect plates. Other fuel cell types, such as PEM, molten carbonate, phosphoric acid, etc. may also be used.

(18) The modular fuel cell system enclosure **10** also contains one or more input or fuel processing modules **16**. This module **16** includes a cabinet which contains the components used for pre-processing of fuel, such as desulfurizer beds. The fuel processing modules **16** may be designed to process different types of fuel. For example, a diesel fuel processing module, a natural gas fuel processing module, and an ethanol fuel processing module may be provided in the same or in separate cabinets. A different bed composition tailored for a particular fuel may be provided in each module. The processing module(s) **16** may processes at least one of the following fuels selected from natural gas provided from a pipeline, compressed natural gas, methane, propane, liquid petroleum gas, gasoline, diesel, home heating oil, kerosene, JP-5, JP-8, aviation fuel, hydrogen, ammonia, ethanol, methanol, syn-gas, bio-gas, bio-diesel and other suitable hydrocarbon or hydrogen containing fuels. If desired, a reformer **17** may be located in the fuel processing module **16**. Alternatively, if it is desirable to thermally integrate the reformer **17** with the fuel cell stack(s), then a separate reformer **17** may be located in each hot box **13** in a respective power module **12**. Furthermore, if internally reforming fuel cells are used, then an external reformer **17** may be omitted entirely.

(19) The modular fuel cell system enclosure **10** also contains one or more power conditioning modules **18**. The power conditioning module **18** includes a cabinet which contains the components for converting the fuel cell stack generated DC power to AC power, electrical connectors for AC power output to the grid, circuits for managing electrical transients, a system controller (e.g., a computer or dedicated control logic device or circuit). The power conditioning module **18** may be designed to convert DC power from the fuel cell modules to different AC voltages and frequencies. Designs for 208V, 60 Hz; 480V, 60 Hz; 415V, 50 Hz and other common voltages and frequencies

may be provided.

(20) The fuel processing module **16** and the power conditioning module **18** may be housed in one input/output cabinet **14**. If a single input/output cabinet **14** is provided, then modules **16** and **18** may be located vertically (e.g., power conditioning module **18** components above the fuel processing module **16** desulfurizer canisters/beds) or side by side in the cabinet **14**.

(21) As shown in one exemplary embodiment in FIG. **1** one input/output cabinet **14** is provided for one row of six power modules **12**, which are arranged linearly side to side on one side of the input/output module **14**. The row of modules may be positioned, for example, adjacent to a building for which the system provides power (e.g., with the backs of the cabinets of the modules facing the building wall). While one row of power modules **12** is shown, the system may comprise more than one row of modules **12**. For example, as noted above, the system may comprise two rows of power modules stacked back to back.

(22) Each of the power modules **12** and input/output modules **14** include a door **30** (e.g., hatch, access panel, etc.) to allow the internal components of the module to be accessed (e.g., for maintenance, repair, replacement, etc.). According to one embodiment, the modules **12** and **14** are arranged in a linear array that has doors **30** only on one face of each cabinet, allowing a continuous row of systems to be installed abutted against each other at the ends. In this way, the size and capacity of the fuel cell enclosure **10** can be adjusted with additional modules **12** or **14** and bases **20** with minimal rearranging needed for existing modules **12** and **14** and bases **20**. If desired, the door **30** to module **14** may be on the side rather than on the front of the cabinet.

(23) FIG. **2** illustrates a plan view of a fuel cell system hotbox **13** including a fuel cell stack or column **40**. The hotbox **13** is shown to include the fuel cell stack or column **40**. However, the hotbox **13** may include two or more of the stacks or columns **40**. The stack or column **40** may include the electrically connected fuel cells **45** stacked on one another, with the interconnects **50** disposed between the fuel cells **45**. The first and last fuel cells **45** in the stack or column are disposed between an respective end plate **60** and interconnect **50**. The end plates **60** are electrically connected to electrical outputs of the fuel cell stack or column **40**. The hotbox **13** may include other components, such as fuel conduits, air conduits, seals, electrical contacts, etc., and may be incorporated into a fuel cell system including balance of plant components. The fuel cells **45** may be solid oxide fuel cells containing a ceramic electrolyte, such as yttria stabilized zirconia (YSZ) or scandia stabilized zirconia (SSZ), an anode electrode, such as a nickel-YSZ, a Ni-SSZ or a nickel-samaria doped ceria (SDC) cermet, and a cathode electrode, such as lanthanum strontium manganite (LSM)). The interconnects **50** and/or end plates **60** may comprise any suitable gas impermeable and electrically conductive material, such as a chromium—iron alloy, such as an alloy containing 4 to 6 wt % iron and balance chromium. The interconnects **50** electrically connect adjacent fuel cells **45** and provide channels for fuel and air to reach the fuel cells **45**.

(24) FIG. **3A** shows a fuel cell power generation system **300** according to embodiments. As shown in FIG. **3A**, the system **300** may include an Input/Output Module (IOM) **310**, a fuel cell system **320**, and an Auxiliary Output Module (AOM) **330**. The system **300** may be electrically connected to an uninterruptible power system (UPS) **340**, a load **350**, and a Resistive Load Bank (RLB) **360** through an optional multi-way switch **370**. A DC power bus **380** connects the SOFC system **320** to the UPS **340** through the switch **370**. An auxiliary DC power bus **390** connects the RLB **360** through the switch **370** to the DC power bus (or to the UPS **340**). However, it is to be understood that embodiments described herein do not require every one of components **310-390**.

(25) IOM **310** may connect SOFC system **320** to the power grid **301**. IOM **310** can include inverter **310A**, as shown FIG. **3A**, for converting a DC output from SOFC system **320** to AC for use by the power grid **301**. IOM **310** can also include other suitable components (not shown) including, but not limited to, controllers, resistive load banks, circuit breakers, and relays. It is to be understood that IOM **310** is optional, as is connecting the fuel cell system **320** the grid **301**. It is to be further understood that the features or components of IOM **310** may be incorporated into other

components, such as fuel cell system **320**.

(26) The fuel cell system **320** is shown in FIG. 3A to include a generalized power source **320a** for the sake of convenience. The fuel cell system **320** and/or power source **320a** may include, for example, the SOFC system shown in FIG. 1 which contains the hot box **13** shown in FIGS. 1 and 2. For sake of brevity, the fuel cell system **320** is referred to below as a “SOFC system **320**”. However, it should be understood that the fuel cell system **320** may include other types of fuel cells, such as PEM fuel cells, molten carbonate fuel cells, etc. The SOFC system **320** may also include a number of other suitable components (not shown), such as energy storage devices (e.g., batteries or supercapacitors), fuel valves, fuel and air blowers, circuit breakers, temperature gauges, etc.

(27) SOFC system **320** may include a controller **320b** as shown in FIG. 3A. Controller **320b** may include any suitable logic that can control certain aspects of SOFC system **320** and/or power source **320a**. For example, controller **320b** may control the fuel flow rate into the SOFC system, the output voltage from the SOFC system, air flow rate into the SOFC system, and fuel recycling rate in the SOFC system, etc. For example, the controller **320b** may control the output voltage and/or power of SOFC system **320** to other components of the system **300**, such as to IOM **310** and/or AOM **330**. In addition, controller **320b** may communicate, either directly or indirectly, with other components in the system **300** and/or with remote control terminals. As discussed in more detail below, controller **320b** can allow SOFC system **320** to indirectly communicate with other components, such as the UPS **340**, by drooping the output voltage of the SOFC system **320**.

(28) AOM **330** may include a DC/DC converter **330a** for providing DC voltage output from the SOFC system **320** at a higher and/or lower voltage compatible with the UPS **340**. Also, as shown in FIG. 3A, one output of the AOM **330** may go to RLB **360**. Although FIG. 3A shows a single connection between AOM **330** and both UPS **340** and RLB **360**, it is to be understood that each of UPS **340** and RLB **360** may be connected separately to SOFC system **320**. As shown in FIG. 3A, AOM **330** may also include a controller **330b**. Controller **330b** may be similar to controller **320b**, as described above in the context of SOFC system **320**, and may perform some similar functions as controller **320b**, as appropriate for AOM **330**. Alternatively, a single controller may control both the SOFC system **320** and the AOM **330**.

(29) UPS **340** may be configured to receive the DC input from the AOM **330**. Alternatively UPS **340** may be configured to receive a DC output from SOFC system **320** directly. UPS **340** may include a DC/DC converter **340a** and inverter **340b**. Converter **340a** may step up and/or step down the DC output voltage received from either AOM **330** or SOFC system **320**. As shown in FIG. 3A, UPS **340** may provide the output of DC/DC converter **340a** to an energy storage device **340c**, such as a bank of supercapacitors, batteries, etc. The energy storage device **340c** is referred to as a “battery” herein for brevity. The battery **340c** may store the output from either AOM **330** or SOFC system **320** further use. The battery **340c** may also supply power to load **350** via inverter **340b** when, for example, SOFC system **320** is off-line or is outputting power lower than the requirements for load **350**. Inverter **340b** may convert a DC output from other parts of UPS **340** to AC output, so that it may be utilized by the AC load **350**.

(30) As shown in FIG. 3A, UPS **340** may also include a controller **340d**. Controller **340d** may be similar to controllers **320b** and **330b**, as described above in the context of SOFC system **320** and AOM **330** above, and may perform some similar functions as controllers **320b** and **330b**, as appropriate for UPS **340**. In addition, as discussed in more detail below, controller **340d** may communicate indirectly with other components, such as controller **320b**, via parameters such as an output voltage of SOFC system **320**. In one embodiment, the controller **340d** does not directly communicate with the controllers **320b** and/or **330b**. For example, the controller **340d** may be made by a different manufacturer than controller **320b** and/or **330b** such that they lack a common communication protocol or a direct data transfer wired or wireless connection.

(31) Load **350** may include a number of suitable electronic devices that can draw power from an

AC power supply, such as the UPS **340**. Load **350** may include a “micro-grid,” in the sense that load **350** may include a network of devices drawing power from fuel cell system **300** through the UPS **340**. For example load **350** may include various computers or and/or servers, such as servers arranged in a server farm, electromechanical devices, lighting fixtures, climate control systems, etc. Generally, load **350** includes devices meant to draw power from SOFC system **320**. Load **350** may also be optionally separately connected to other components (including the grid **301**, alternative power supplies, batteries, etc.) so that it may draw power when SOFC system **320** is providing less power than required by load **350**. In addition, as discussed above, load **350** may draw power from battery **340c** of UPS **340** when SOFC system **320** is providing less power than required by load **350**. In one embodiment, when UPS **340** is a microgrid AC power supply and controller which operates in an islanded configuration (i.e., where the UPS **340** and load **350** are not connected to the grid **301**).

(32) RLB **360** may include a number of suitable impedance imparting devices, including resistors, inductors, capacitors, and/or energy storage devices such as batteries or supercapacitors. For example, RLB **360** may include a number of resistive elements **360a** shunted to ground **360b**. In this or similar configurations, RLB **360** may allow the shunting to ground **360b** of DC power produced by SOFC system **320** when the power demand by load **350** is less than the output power of SOFC system **320**. This may allow SOFC system **320** operate at higher power and, therefore, higher for a more efficient and robust operation. This potential use of RLB **360** will be explained in more detail below.

(33) As shown in FIG. 3A, RLB **360** may further include an energy storage device **360c**, such as a battery or supercapacitor, which is referred to as a “battery **360c**” for brevity herein. One purpose for battery **360c** may be to store power delivered to RLB **360** via AOM **330**. Power stored by battery **360c** may be used, for example, to supplement power supplied to UPS **340** by SOFC system **320** when a power demand from load **350** exceeds a certain threshold, such as the maximum output power that can be provided by SOFC system **320** at any given time. In addition, as explained in more detail below, battery **360c** may store power produced by SOFC system **320** when the output power exceeds a power demand by load **350**.

(34) An optional multi-way switch **370** may be located on the DC power bus **380** between the AOM **330**, UPS **340**, and RLB **360**. The switch **370** may include switching elements, such as transistor(s), relay(s), contactor(s), etc. and an optional controller (i.e., logic). Alternatively, the switch **370** may be controlled by the UPS **340** controller **340d**. The switch **370** may control current flow between the AOM **330** and one or both of UPS **340** and RLB **360**, and current flow between the UPS **340** and the RLB **360**.

(35) FIG. 3B shows steps in an exemplary method for controlling the output power of the system **300** shown in FIG. 3A. In step **361**, output power is provided from a power source **320a** to a load **350** to track and meet a power demand of the load. Any suitable power source may be used. For example, the power source **320a** may be the SOFC system **320** which provides power to one or more devices in load **350** in order to meet a load power demand. As will be explained in more detail below, other devices may be relied upon to supply power at various steps including step **361**. These other devices include, but are not limited to, batteries **340c** and/or **360c**. In any case, the power source **320a** provides power in step **361** such that the power demand from load **350** is substantially met. In this step, as explained in more detail below, excess power may also be routed to power storage devices such as batteries **340c** and/or **360c** and/or the resistive elements **360a** of the resistive load bank **360**. Power stored in batteries **340c** and/or **360c** may be later used to supply power to load **350**.

(36) In step **362**, in response to an increased power demand from the load **350**, it is determined if the requested increase in output power of the power source **320a** (e.g., of the SOFC system **320**) which equals to the increased power demand would cause the output power of the power source to exceed a threshold output power. The threshold output power can be a maximum output power

above which SOFC system **320** is not capable of providing more power at a given time (e.g., due to temporary lack of sufficient fuel flow or power which is above the maximum rated power of the SOFC system) or an output power above which the SOFC system **320** becomes less efficient or could become damaged. In one embodiment, the determination that the output power of the SOFC system **320** would reach the threshold output power is made by the SOFC system controller **320b**. Other operating conditions that may be optionally detected and determined in step **362** include the temperature at which SOFC system **320** operates and/or the fuel flow rate into the SOFC system **320**.

(37) If it is determined that the requested increase in output power of the power source **320a** (e.g., of the SOFC system **320**) would exceed the threshold output power, then in step **363**, the power source generates a signal that the threshold output power would be exceeded by changing an output voltage of the power source. For example, the SOFC system controller **320b** changes the output voltage of the SOFC system **320** on the DC power bus **380** to signal to the UPS **340** controller **340d** that the output power of the SOFC system **320** would exceed the threshold output power. For example, the signal can be a small decrease or increase in the output voltage of SOFC system **320** (a “droop”), rather than a direct communication between the SOFC system **320** controller **320b** and UPS **340** controller **340d**. For example, the droop can be a less than a 5% decrease or increase in output voltage, such as a 0.5 to 4.0%, including a 1 to 3%, decrease or increase in output voltage. Signaling using an output voltage can easily be detected by components downstream from SOFC system **320**, such as the UPS **340** controller **340d**, and has the advantage in that the signal can be detected and understood by a variety of different downstream components. In other words, no specific communication protocol or hardware need be used in order to implement the signal as the output voltage droop. As described in more detail below, this allows SOFC system **320** to communicate with components, such as the UPS **340**, that have not been especially made or configured to directly communicate with SOFC system **320**.

(38) In step **364**, the output power from power source **320a** is not increased above the threshold output power such that the output power no longer tracks the power demand of the load **350** if it was determined in step **362** that the threshold output power would be exceeded. In this step **364**, the SOFC system controller **320b** stops increasing the output power of the SOFC system **320**. The termination in the increase in output power of the SOFC system **320** above the threshold output power in step **364** results in the output power no longer tracking the power demand of load **350**. For example, the output power may be set in step **364** to a constant value that equals the threshold output power of SOFC system **320**. As discussed in more detail below, this output power setting may enable the SOFC system **320** to operate more efficiently and with less potential for long-term damage. The excess power to satisfy the load **350** power demand can be provided from the energy storage devices **340c** and/or **360c** (e.g., in an islanded or grid connected configuration) and/or from the grid **301** (e.g., in the grid connected configuration) by the UPS controller **340d** in response to receiving the signal in step **363**. For example, in response to receiving the signal from the SOFC controller **320b** (i.e., the change in output voltage from the SOFC system **320**), the UPS controller **340d** may change the setting of the switch **370** to allow additional power (i.e., current) to flow from the batteries **340c** and/or **360c** through the UPS **340** to the load **350**.

(39) In contrast, if it is determined in step **362** that the increase in output power of the power source **320a** (e.g., of the SOFC system **320**) would not exceed a threshold output power, then the output power from the power source is increased to meet the power demand of the load **350**. For example, if the UPS controller **340d** does not detect the change in output voltage of the SOFC system **320** (i.e., if the SOFC system controller **320b** determines that meeting the load power demand would not exceed the threshold output voltage and does not generate the signal), then the SOFC controller **320b** increases the output power of the SOFC system **320** to meet the load power demand and the UPS controller **340d** draws the increased output power from the SOFC system **320** to track and meet the load **350** power demand.

(40) FIGS. 4A-4D show certain aspects of system 300 in operation according to embodiments. Specifically, FIG. 4A shows the load power demand (in Watts) 410 exerted by load 350 on SOFC system 320 versus time. FIG. 4B shows the power output (in Watts) from SOFC system 320 over the same time period. FIG. 4C shows the output voltage of SOFC system 320 over the same time period.

(41) Both FIGS. 4A and 4B show the load 350 power demand 410 provided by the load 350 to the SOFC system 320. For example, in the embodiment shown in FIGS. 4A and 4B, threshold output power 420 may be, for example, a threshold indicative of a user imposed constraints or operating conditions. For example, threshold output power 420 may correspond to the highest power at which a desired efficiency and/or operating temperature of the SOFC system 320 is achieved and/or the highest power at which SOFC system 320 damage is avoided. As another example, threshold output power 420 may relate to an output power corresponding to a desired fuel utilization by SOFC system 320. Alternatively, as will be described in more detail below with respect to additional embodiments, the threshold output power 420 may be, for example, a maximum possible output power of the SOFC system 320 at a given time. For example, the threshold output power 420 may be the maximum output power of the SOFC system when the load power demand 410 increases at a high rate such that the SOFC system temporarily cannot meet the increased load power demand 410 because the SOFC system 320 cannot ramp up the output power 430 at the corresponding sufficiently high rate. For example, the SOFC system 320 may not be able to ramp up the output power 430 at the sufficiently high rate due to temporary lack of sufficient fuel flow, since it takes a certain amount of time to sufficiently increase the fuel flow rate to the SOFC system 320 to achieve the output power 430 demanded by the load 350. Alternatively, the threshold output power 420 may be the maximum rated power of the SOFC system 320.

(42) As shown in FIG. 4B, SOFC system 320 output power 430 reaches threshold output power 420 at a time t1. Before time t1, i.e., between time t0 and time t1, SOFC system 320 output power 430 tracks and meets the load 350 power demand 410. In other words, SOFC system 320 supplies all the power according to the requirements of load 350. FIGS. 4A and 4B show load power demand 410 continues to increase beyond threshold output power 420 from time t1 to time t2. However, once the load 350 power demand 410 reaches the threshold output power 420, the SOFC system 320 may be operating at a maximum safe, desired and/or possible output power. Although SOFC system 320 may be configured to output even more power than threshold output power 420, doing so may damage the system 320 and/or degrade its efficiency. Therefore, at time t1, it may be advantageous to prevent or stop the output power 430 from increasing even though, as shown in FIG. 4A, the load power demand 410 continues to increase.

(43) One response, as shown in FIG. 4B, is to maintain the output power 430 of SOFC system 320 at the threshold output power 420 from time t1 to time t2, regardless of the increase in load power demand 410. Maintaining this condition supplies load 350 with the threshold output power 420. The output power 430 of the SOFC system may be maintained at the threshold output power 420 by the SOFC system controller 320b to stop the output power 430 from following the increasing load power demand 410. In other words, if the output power 430 was already equal to the threshold output power 420 before the increase in the load power demand 410, then the SOFC system controller 320b does not increase the output power 430. If the output power 430 was below the threshold output power 420 before the increase in the load power demand 410, then the SOFC system controller 320b only increases the output power 430 to equal the threshold output power 420, but does not increase the output power 430 above the threshold output power 420.

(44) FIG. 4C shows that the SOFC controller 320b changes the output voltage 440 of the SOFC system 320 to signal to the UPS controller 340d that the threshold output power 420 would be exceeded if the load power demand 410 is met. As shown in FIG. 4C, the output voltage 440 of SOFC system 320 is maintained at an initial voltage V1 from time t0 until time t1, when the output power 430 reaches threshold output power 420. At time t1, SOFC system 320 signals that threshold

output power **420** would be exceeded by slightly decreasing output voltage **440**. This decrease may be called a “voltage droop” **450** and may occur over time Δt (e.g., over 0.1 seconds to 10 seconds), as shown in FIG. 4C. Droop **450** can be detected by UPS **340** and/or by other components. In particular, UPS **340** controller **340d** may detect the droop **450** and may respond to the detection of droop **450** by leveling off the power output **430** it receives from SOFC system **320**, thus applying less power from the SOFC system **320** to load **350** than the load power demand **410**. In other words, the droop **450** signals reaching and maintaining output power **430** at threshold output power **420**, potentially causing other components of system **300** to supply supplemental power to UPS **340**.

(45) As shown in FIG. 4C, droop **450** is generally the difference between a steady state voltage V_1 output by SOFC system **320** from time t_0 to time t_1 and a lower voltage V_2 applied from t_1 onward. Droop **450** may be fixed percentage of V_1 , for example less than 5% of V_1 , such as 0.5% to 4%, or 1% to 3% of V_1 . Generally, it can be advantageous for droop **450** to represent a large enough change in output voltage **440** to be detected by controller **340d**. At the same time, it may be advantageous for droop **450** to be small enough in magnitude to be acceptable by the load **350** operating specification and to result in relatively insubstantial change to the output power **430** of SOFC system **320** such that the operation of other components (e.g., UPS **340** or devices in load **350**) are not substantially affected by the change in output voltage. In some embodiments it may be advantageous for droop **450** to be substantially less than 5% of V_1 (e.g., 3% or less). Droop **450** can be chosen to best fit the particular application and the sensitivity of the various components involved (e.g., UPS **340** controller **340d**). Droop **450** can alternatively be chosen for more general applications allowing increased flexibility to use SOFC system **320** with a variety of different components, such as UPS **340**, with different specifications and made by different manufacturers without the express aim of being compatible with SOFC system **320**.

(46) Advantages of using a voltage droop **450** over other means of communication (e.g., wired or wireless communication) between SOFC system **320** and UPS **340**, include that it does not require particular circuitry, connection, or communication protocol between controller **320b** and controller **340d**. In other words, communication by voltage droop **450** does not require special hardware or communication protocols to allow controller **320b** to communicate a change in output power **430** to other components, including controller **340d**. Instead, controller **340d** need only be configured such that it interprets droop **450** as a signal indicative of changing power output. It is to be understood that droop **450** may signal other changes in the performance or operation of the system **300** than a change in output power **430**, as shown in FIG. 4B. For example, droop **450** may signal the controller **340d** to trigger battery **360c** and/or **340d** to supply supplemental power to load **350**, and for the switch **370** to provide a current path between the battery **360c** and the load **350**.

(47) FIG. 4D shows use of droop **450** to optionally trigger at least one energy storage device (e.g., battery **340c** and/or **360c**) to supply supplemental power **460** to load **350**. In the example shown in FIG. 4D, the output power of the one or more battery supplements the output power from SOFC system **320** so that the total power supplied to the load **350** (i.e., the sum of the power supplied by at least one battery and SOFC system **320**) is substantially equal to the load power demand **410**. It is to be understood that this arrangement is merely exemplary. Droop **450** may trigger supplemental power from the least one battery of any magnitude. For example, droop **450** may trigger supplemental power from the at least one battery that is proportional to power output **430** of SOFC system **320** or to load power demand **410**. Although FIG. 4D shows a single droop **450** effecting a single change in the supplemental power **460**, it is to be understood that multiple droops are possible. In some arrangements, it may be advantageous to use multiple droops to trigger a stepwise change in the supplemental power **460** output by the at least one battery **340c** and/or **360c**. Although FIG. 4D shows control of one or more battery via droop **450**, it is to be understood that droop **450** may be used in the way shown to control the provision of supplemental power from any other suitable supplemental power source (e.g., the grid **301**, a supplemental power generator, such

as a diesel generator, etc.).

(48) The UPS **340** can be operated in three modes. In the first mode, the UPS **340** controller stops increasing power to the load **350** when it detects a droop **450** of any magnitude. In the second mode, the SOFC system controller **320b** can output a gradient droop (i.e., a varying magnitude droop) **450** to signal that the SOFC system **320** can provide an output power **430** increase to the UPS **340** and load **350** that is inversely proportional to the absolute magnitude of the droop **450**. In this mode, the gradient droop **450** is used to signal the UPS controller **340d** that the SOFC system **320** is either approaching the threshold output power **420** or that the SOFC system **320** cannot keep up with the increase in load power demand **410**. The UPS **340** provides the increased output power **430** to the load **350** that does not exceed the threshold output power **420** and that only partially meets the load power demand **410**. In the third mode, the UPS controller **340d** provides a given amount of supplemental power from one or more batteries **340c** and/or **360c** to the load **350** based on the magnitude of the gradient droop **450** described above for the second mode. For example, based on the larger or smaller the absolute magnitude of the droop **450** detected by the UPS controller **340d**, the UPS controller **340d** applies a respective larger or smaller amount of supplemental power **460** output from the at least one battery to the load **350** to completely meet the load power demand **410**.

(49) Still other uses for droop **450** may be implemented. For example, droop **450** may alternatively indicate to controller **340d** that interruption in power from SOFC system **320** is imminent. Droop **450** may alternatively also indicate to controller **340d** that a decrease in power from SOFC system **320** is imminent. Controlling the UPS **340** based on droop **450** allows the UPS **340** to draw supplemental power **460** as long as the controller **340d** can issue a command based on the significance of droop **450**. The significance or meaning of droop **450** may also be altered by a user or administrator by reprogramming controller **340d**. In this way, the signaling via droop **450** shown in FIG. **4C** allows SOFC system **320** to be used with a variety inverters, such as the UPS **340** that are not configured to be in direct data and information communication with the SOFC system **320**. It is to be understood that other variations in the information conveyed by droop **450** are encompassed by the embodiments described herein.

(50) FIG. **5** illustrates an alternative embodiment in which the threshold output power **430** may be the maximum power output of the SOFC system **320**, such as when the load power demand **410** increases at a high rate such that the SOFC system temporarily cannot meet the increased load power demand **410** because the SOFC system **320** cannot ramp up the output power **430** at the corresponding sufficiently high rate. For example, as described above, the SOFC system **320** may not be able to ramp up the output power **430** at the sufficiently high rate due to temporary lack of sufficient fuel flow, since it takes a certain amount of time to sufficiently increase the fuel flow rate to the SOFC system **320** to achieve the output power **430** demanded by the load **350**. The y-axis in FIG. **5** is a magnitude of power or voltage, while the x-axis is time.

(51) In the embodiment shown in FIG. **5**, at time **t1**, the load **350** power demand **410** experiences a high rate spike from power **P1** to power **P2**. In response, the SOFC system **320** begins to increase the output power **430** at time **t1**. However, the output power **430** is ramped up gradually (e.g., due to the delay in increasing the fuel flow rate and/or other operating characteristics of the SOFC system **320**) from initial output power **P1** to demanded power **P2**. Thus, the output power **430** does not reach the demanded power **P2** until time **t4**. At the same time **t1**, the SOFC system **320** controller **320b** sends a signal to the UPS controller **340d** that the SOFC system **320** output power **430** cannot track and meet the load power demand **410** at this time **t1**. In other words, the controller **320b** decreases the output voltage **440** provided by the SOFC system **320** on the DC power bus **380** connecting the SOFC system **320** and the UPS **340** to signal that the SOFC system **320** cannot provide output power which equals to load power demand value **P2**.

(52) As shown in FIG. **5**, the controller **320b** may provide a gradient droop **450** to the UPS **340** which gradually decreases to signal that the SOFC system **320** output power **430** will be increasing

gradually at a corresponding rate. For example, the output voltage may decrease from a steady stage voltage value **V1** to a lower value. After decreasing at a given rate (e.g., slope) from time **t1** to time **t2** (which is less than time **t4**), the output voltage **440** may level off at a constant value **V2** at time **t2**. The constant value **V2** of the output voltage **440** is lower than value **V1**, but is higher than the acceptable minimum DC voltage value **V3** that is specified for the DC power bus **380**. For example, $V3 > 0.95 \cdot V1$, and $V1 > V2 > V3$. Thus, the droop **450** changes from gradient value (i.e., variable value between **V1** and **V2**) to a constant value **V2** at time **t2**. The slope and maximum value of the gradient droop may indicate the specific value of the supplemental power **460** that should be provided to supplement the SOFC system **320** output power **430** over a given time to meet the load power demand **410**. The start of constant value of the droop **450** at time **t2** may signal the UPS controller **430d** that the information about the output power **430** ramp is completed, and that the UPS controller **430d** should begin drawing the supplemental power **460** at time **t2**. Thus, the system operates in the third mode described above.

(53) In response to receiving the signal (i.e., detecting the droop **450** on the DC power bus **380**), the UPS controller **430d** begins at time **t2** to draw supplemental power **460** from at least one energy storage device (e.g., battery **440c** and/or **460c**) and/or from another source, such as from the grid **301** and/or from a backup power source, such as a generator. As shown in FIG. 5, the initial value of the supplemental power **460** drawn is spiked from zero kW to a value of **P2** minus **P1** to meet the load power demand **410** at time **t2**. The amount of supplemental power **460** drawn by the UPS **340** is inversely proportional to the amount of output power **430** from the SOFC system **320**. In other words, as the output power **430** from the SOFC system **320** increases from value **P1** to value **P2** (which is reached at time **t4**), the supplemental power **460** decreases from value (**P2-P1**) at time **t2** to an intermediate value at time **t3** to zero kW at time **t4**.

(54) Once the output power **430** of the SOFC system **320** reaches value **P2** of the power demand **410** at time **t4**, the SOFC system controller **320b** terminates the droop **450**. In other words, the controller **320b** increases the output voltage **440** of the SOFC system **320** from the constant droop value **V2** back to the initial value **V1**. Upon detecting the termination of the droop **450**, the UPS controller **430d** stops drawing the supplemental power **460** at time **t4**. After time **t4**, the output power **430** from the SOFC system **320** tracks and meets the load **350** power demand **410** by supplying the demanded power value **P2** to the load **350** through the UPS **340**.

(55) Thus, in this embodiment, the signaling between the power source **320a** and the downstream device (e.g., UPS **340**) is managed by changing the voltage output of the power source **320a**. The distribution of power generated by the SOFC system **320** on the DC power bus **380** between the DC output and the AC output can be controlled in an islanded or grid parallel operation. Furthermore, the output power delivered to the DC power bus **380** can be controlled redundantly through the various inverters **310a** and **340a** by adding resistive loads **360a** on the DC output of the SOFC system **320**.

(56) FIGS. 6A-6C illustrate another embodiment in which the output power **730** of a power source **320a**, such as the SOFC system or another power source described above, is maintained at or above a minimum desired operating power **720** even after the load **350** power demand **710** drops below the minimum desired operating power **720**. The minimum desired operating power **720** may correspond to an output power below which the temperature of the SOFC system **320** becomes so low that the output efficiency of SOFC system **320** is degraded. That is, below the minimum desired operating power **720**, the efficiency of SOFC system **320** is less than desired, for example, is at least 20% below optimal efficiency. Operating SOFC system **320** at such temperatures may cause damage to SOFC system **320** degrading its long-term performance. Therefore, it may be advantageous to maintain the output power **730** of SOFC system **320** above minimum desired operating power **720** even after the load **350** power demand **710** drops below the minimum desired operating power **720**.

(57) In one embodiment shown in FIG. 6A, the load **350** power demand **710** may be above the

minimum desired operating power **720** and below the threshold output power **420** of the SOFC system **320**. For example, the load **350** power demand **710** may reach a maximum value at time $t_{sub.max}$ and then decrease. The load power demand **710** may drop below the minimum desired operating power **720** of SOFC system **320** at time t_1 .

(58) In response to the load power demand **710** dropping below the minimum operation power **720** at time t_1 , a portion or all of the output power **730** of the SOFC system **320** is directed to the RLB **360** through the auxiliary DC power bus **390** to maintain the total output power **730** of the SOFC system **320** at or above the minimum desired operating power **720**, as shown in FIG. 6B. For example, if the load power demand **710** drops to zero, then all of the output power **730** is directed to the RLB **360** through bus **390**. If the load power demand **710** drops to a value above zero and below the minimum desired operating power **720**, then a portion of the output power **730** is directed to the load **350** through the DC power bus **380** and the UPS **340** to meet and track the load power demand **710**, while an excess portion of the output power **730** is directed to the RLB **360** via the auxiliary DC power bus **390**.

(59) The output power **730** may be split between the UPS **430** and the RLB **460** by the switch **370**. The switch **370** may be controlled by any suitable controller, such as the SOFC system **320** controller **320b**, the UPS **430** controller **430d** and/or a dedicated switch **370** controller.

(60) In one embodiment shown in FIG. 6C, if the switch **730** is controlled by the UPS **430** controller **430d** and/or the dedicated switch **370** controller, then the SOFC system **320** controller **320b** can change the output voltage **440** (i.e., to create a voltage droop **450**) on the DC power bus **380** to signal the UPS **430** controller **430d** and/or the dedicated switch **370** controller to direct all or a portion of the output power **730** to the RLB **360**. Similar to the prior embodiment, a gradient droop **450** may be used and the slope and maximum value of the gradient droop may indicate the specific value or portion of the total output power **730** that should be directed to the RLB **360** by the switch **370** via bus **390**.

(61) Although FIG. 6B shows the output power **730** of SOFC system **320** ramping up to its threshold output power **420**, it is to be understood that the output power may be maintained at other levels between the minimum desired operating power **720** and the threshold output power **420** in this embodiment. For example, it may be advantageous to increase output power **730** from SOFC system **320** to a level at which the SOFC system **320** is operating more efficiently than it would be at the minimum desired operating power **720**. Generally, this output power **730** corresponds to a relatively high operating temperature of SOFC system **320** (e.g., above 700°C ., such as 800 to 850°C . as discussed above). Thus, in this embodiment, the SOFC system **320** temperature is maintained in the desired operating range (e.g., above 700°C ., such as 800 to 850°C .) by maintaining the output power **730** above the load power demand **710** and directing the excess portion of the output power **730** that is higher than the load power demand **710** to the RLB **360**. Thus, the direction of power to the RLS **360** can be managed such that the load steps on the SOFC system **320** can be minimized, and a sufficient load on the SOFC system **320** can be maintained to maintain the SOFC system **320** within a desired temperature range. Furthermore, the direction of excess power to the RLB **360** manages the overvoltage on the DC power bus **830** by preventing power flow back from the load **350** into the SOFC system **320**.

(62) FIGS. 7A-7C show another configuration according to another embodiment. FIG. 7A shows the load demand **810** as experienced by SOFC system **320** via the UPS **340**. FIG. 7B shows the output power **830** from the SOFC system **320** to the UPS **340**. FIG. 7C shows the excess output power **832** from the SOFC system **320** to the RLB **360**. The total output power **730** shown in FIG. 6B of the previous embodiment corresponds to the sum of the output power **830** to the UPS **340** and the excess output power **832** to the RLB **360**.

(63) As shown in FIGS. 7A and 7B, the output power **830** to UPS **340** tracks and meets the load power demand **810** so that load **350** is fully powered by the SOFC system **320** from time t_0 to time t_2 . FIGS. 7A and 7B also show that from time t_0 to time t_1 , both the load power demand **810** and

the output power **830** to UPS **340** are less than the threshold output power **420** of the SOFC system **320**. Thus, the sum of output powers **830** and **832** is also less than the threshold output power **420**. When the output power **830** increases by a given amount, the excess output power **832** decreases by a corresponding amount to keep the total output power **730** (e.g., the sum of output powers **830** and **832**) of the SOFC system **320** constant with time.

(64) As shown in FIG. 7A, the load power demand **810** increases sharply at **t1** and can for example reach the threshold output power **420** of the SOFC system **320** shortly thereafter. As discussed above with respect to FIG. 5, the SOFC system **320** typically takes a longer time to provide a corresponding increase in output power **830** to the load **350** than the spike in the load power demand **810**. This can leave the load power demand **810** unsatisfied for a period of time. In the prior art, an auxiliary power is typically provided from an energy storage device or the grid to bridge the temporary output power shortage. This involves additional process complexity.

(65) In the present embodiment shown in FIG. 7C, all or a portion of the excess output power **832** that was directed to the RLB **360** is redirected to the load **350** through the UPS **340** to meet the load power demand **810**. Thus, bridging auxiliary power from the energy storage device and/or the grid **301** is not required to meet the spike the load power demand **810**. The UPS controller **340d** may direct the excess output power **832** from the RLB **360** to the load **350** by controlling the switch **370**. The output power **832** to the RLB **360** can be decreased to zero as shown in FIG. 7C or remain above zero, depending on the amount of power needed to meet and track the load power demand **810**.

(66) Although the foregoing refers to particular preferred embodiments, it will be understood that the invention is not so limited. It will occur to those of ordinary skill in the art that various modifications may be made to the disclosed embodiments and that such modifications are intended to be within the scope of the invention. All of the publications, patent applications and patents cited herein are incorporated herein by reference in their entirety.

Claims

1. A power generation system, comprising a power source that is configured to communicate with at least one of a downstream load or a downstream device by changing a voltage on a power bus between the power source and the at least one of the downstream load or the downstream device, while power source provides power on the power bus to the at least one of the downstream load or the downstream device.
2. The power generation system of claim 1, wherein: the power source comprises a DC power source having a power source controller; the power bus comprises a DC power bus; the device comprises an inverter containing device containing an inverter controller; and the inverter controller is a separate device from the power source controller such that there is no separate communication link or communication protocol between the inverter controller and the power source controller.
3. The device of claim 2, wherein: the power source controller is configured to determine if an increase in output power of the power source which equals to an increased power demand from a load would cause the output power of the power source to exceed a threshold output power in response to the increased power demand from the load; the power source controller is configured to communicate to the inverter controller that the threshold output power would be exceeded by changing the output voltage of the power source, and not increasing the output power above the threshold output power such that the output power no longer tracks the power demand of the load if the output power of the power source would exceed the threshold output power; and the power source controller is configured to increase the output power from the power source to meet the power demand of the load if the output power of the power source would not exceed the threshold output power.

4. The device of claim 3, wherein: the load comprises a microgrid; the device comprises an uninterruptable power system containing an energy storage device; and the power source comprises a solid oxide fuel cell (SOFC) system.
